

NOTES

Basic of gears

Gears - Introduction

↳ Drive used in power transmission.

Input

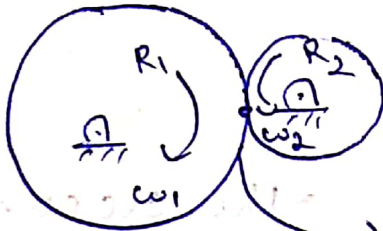
P_1

Gear drive

Output

P_2

Ideally $P_1 = P_2$



$$v = R\omega$$

ω = angular velocity.

Q - pt. of contact.

By Body 1 $\leftarrow v_Q$

By Body 2 $\leftarrow v_Q$

Since ' Q ' is common for both 1 & 2:

$$(v_Q)_1 = (v_Q)_2$$

$$\boxed{\omega_1 R_1 = \omega_2 R_2}$$

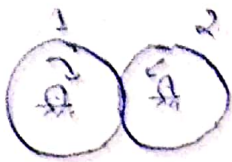
Condition for pure rolling

Hence

$$\boxed{\frac{\omega_1}{\omega_2} = \frac{R_2}{R_1}}$$

Good condition for power transmission

For power transmission



Ideal case

$$\frac{\omega_1}{\omega_2} = \frac{R_2}{R_1}$$

But in reality -

slipping is possible.

eg. $\Rightarrow$ Belt drive, rope drive

$$R_1 \omega_1 \neq R_2 \omega_2 \Rightarrow \frac{\omega_1}{\omega_2} \neq \text{Const}$$

Due to slipping -

1. Friction $\rightarrow$ kinetic

2. Some energy losses.

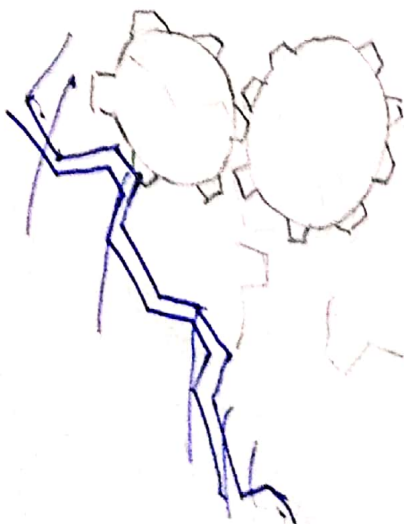
• In some of the machine the accuracy is required in power transmission.

$$\therefore \frac{\omega_1}{\omega_2} = \text{Const.} \quad \underline{\text{Required}}$$

eg. Clock.

$\rightarrow$ The ω ratio is needed in specific ratio.

So; Projection can be made on sprockets so that no slipping occurs



• At least one pair always locked

$\Rightarrow$
Positive drive

Classification of gears

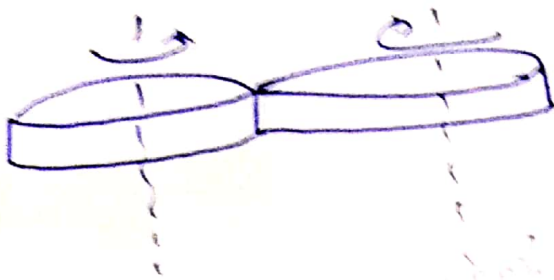
Shaft form the important element of machine. They are element that support rotating part like gears & pulley.

(A) According to the axes of the shaft connected

(B) According to the type of gearing.

(C) According to the axes of shaft connected.

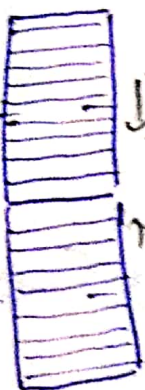
1. Both axis are parallel: Spur
Helical



Single helical
Double Helical.

(a) Teeth are straight and || to the axis of shaft: Spur Gears

used very less nowadays.



Advantage.

→ No axial thrust

Disadvantage:

→ Impact

Instantaneous engagement & disengagement.

For power transmission

• Spur Gears

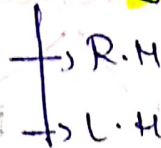
→ used for very low power transmission system.

→ low speed.

→ For high speed it can be used if noise is not a concern.

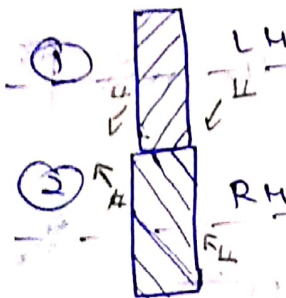
(b) Teeth are straight and inclined to the axes of shaft:

↓
Helical gear (Mostly used)



Advantage

⇒ No/min. Impact stresses
∴ Gradual Engagement & disengagement



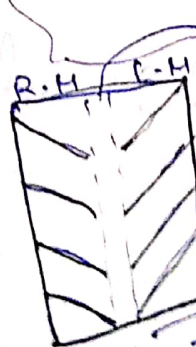
Disadvantage:

⇒ Axial thrust.

→ To overcome this problem.

↓
Double helical gear
or
Herringbone gear

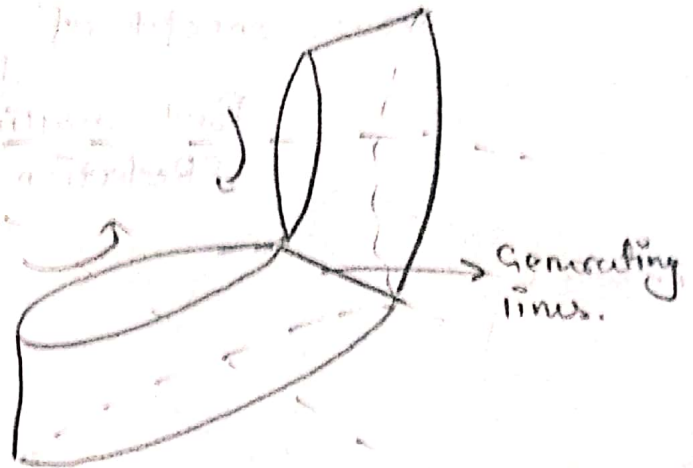
⇒ Cost ↑



Gap → Due to manufacturing limitations.

for cutting tool to change direction.

2. when the axes are intersecting:
(eg.) Bevel gear



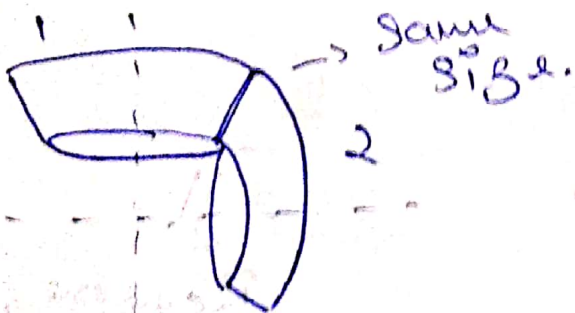
Bevel gears

Straight bevel gear zero bevel gear vertical bevel gear spiral bevel gear.

angle is less

Special case

when axes are $\perp$ to each other.

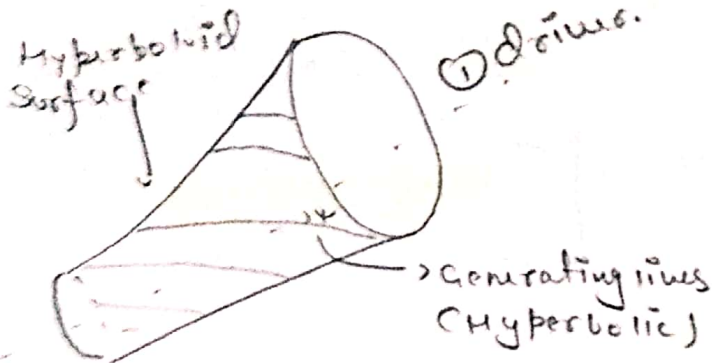


$\frac{\omega_1}{\omega_2} = 1$ { when both gears have same radius and same no. of teeth }
 ✓ Inter gears

3. when the axes are neither $\parallel$ nor intersecting.

$\Downarrow$
New concept of pure rolling is not possible.

$\Downarrow$
But rolling is possible.
(Rotation + sliding)

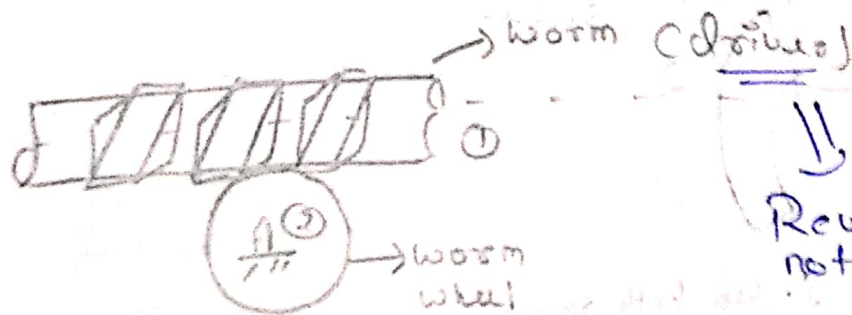


Ψ = spiral angle / helix angle.

[SPIRAL GEAR]

SPIRAL Gear (Curved teeth)

$\Downarrow$
eg: worm and worm wheel
Driver $\Downarrow$ Driven $\rightarrow$ very low spiral angle
 $\downarrow$
very high spiral angle.
very low spiral



$\Downarrow$
Reverse not possible

• very high speed reductions possible.

[B] Based on types of gears.

External

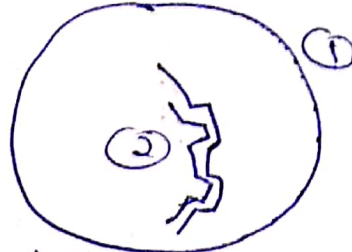
Internal

Rack & Pinion

(1) External



(2) Internal



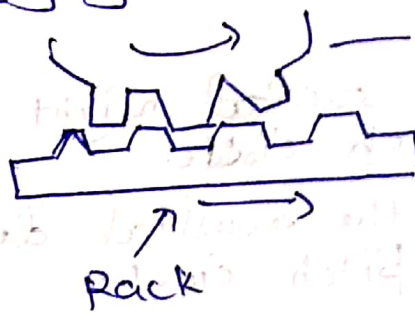
* Limited application $\rightarrow$ Cost $\uparrow$

(3) Racks

straight gears

(Rotation $\leftrightarrow$ Translation)

Application
 $\downarrow$
Rack & pinion
steering
mechanism.



Pinion

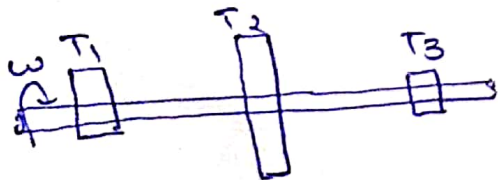
Rack

NOTE:

(1) If more than one gears are mounted on a shaft.

$\rightarrow$ Compound gears.

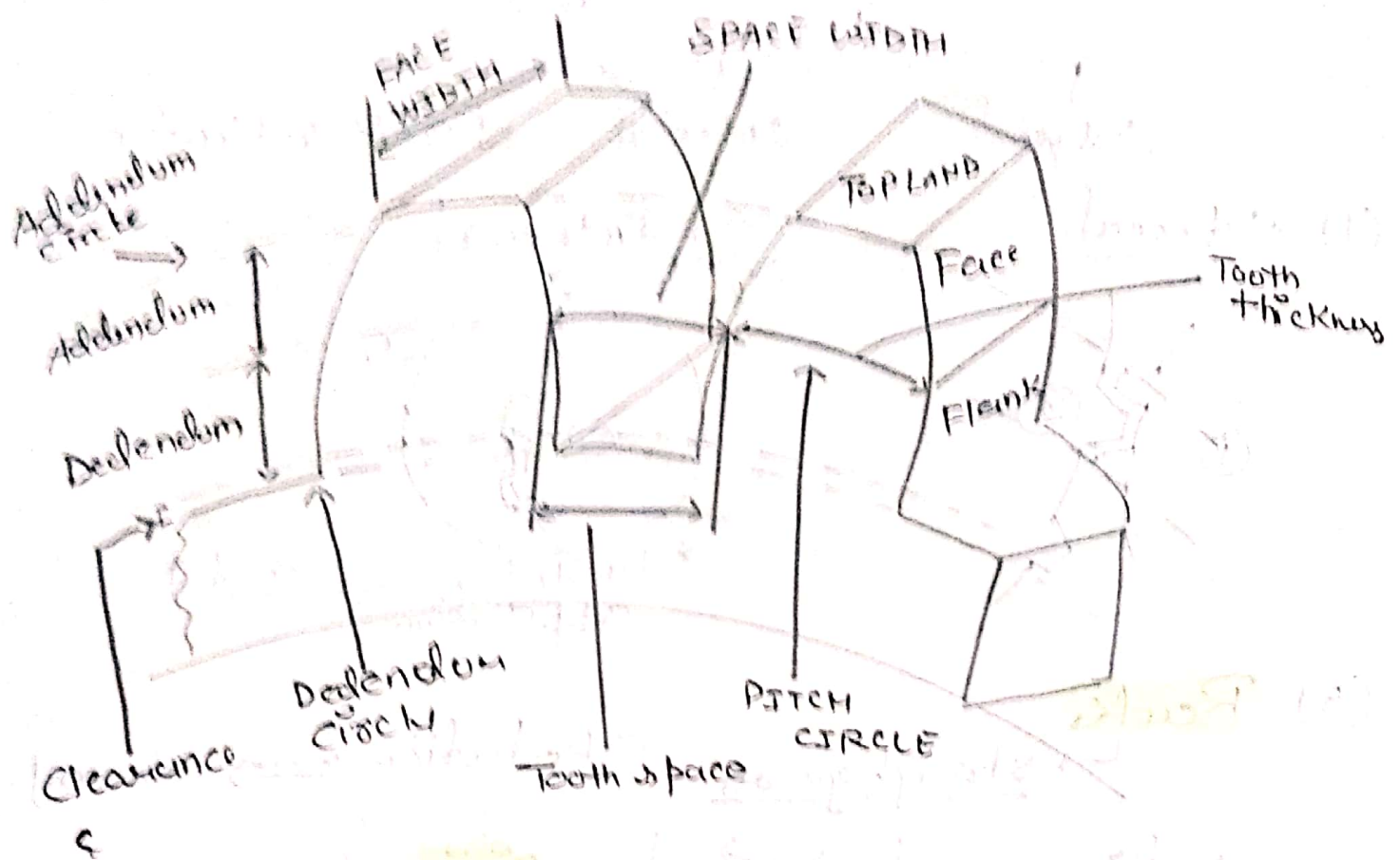
(same speed)



(2) Generally in power transmission, smaller gear are drivers

$$P = T \times \omega$$

Gear terminology



- Addendum - Its the radial height of tooth above the pitch circle.
- Dedendum - Its the radial depth of tooth below the pitch circle.
- Full depth = addendum + Dedendum.